

Vincenzo Barone
INSTM, National Interuniversity Consortium for Materials Science and Technology
via G. Giusti 9
50121 Firenze, Italy
vincebarone52@gmail.com

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Editor
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Dear Editor,

Please consider the manuscript entitled “Algorithmic Generation and Exploitation of Topology-Aware Models for Semiexperimental Structure Refinement” for publication in *The Journal of Chemical Physics*.

The manuscript addresses model formulation, not only model solution. Once a least-squares model has been specified, modern semiexperimental refinement programs can solve it reliably; the harder and less formalized step is the generation of that model. The active coordinates, symmetry restrictions, weakly determined directions, parameter classes, and use of reference structural information still often depend on expert choices, especially for cyclic, fused, and bridged molecular graphs.

We introduce MORPHEUS (Model Optimization and Refinement from Perceived Hierarchy, Equivalence, Uncertainty, and Symmetry), a topology-aware framework that formulates statistically admissible non-redundant refinement models from Cartesian geometry, molecular topology, symmetry, isotopic coverage, and computational reference data. More precisely, the generated models are topology-derived, symmetry-consistent, and rank-consistent: topology defines the primitive coordinate space, symmetry selects the admissible active subspace, isotopic coverage determines which directions are supported, and reference information enters through explicit constraints, predicates, or parameter classes. Two non-redundant, symmetry-adapted coordinate representations are available: generalized internal coordinates (GICs) built from homogeneous primitive blocks, and symmetry-adapted Cartesian displacements with translations and rotations removed.

The work retains the essential ingredients of modern semiexperimental structure refinement, including moment-of-inertia fitting, vibrational and electronic corrections, parameter classes, reduced-dimensionality hydrogen models, covariance propagation, Kraitchman diagnostics, and human-readable reports. Its central contribution is to formalize the construction of the refinement model itself as an algorithm: the active representation and the constraint or parameter-class relations are topology-derived, symmetry-consistent, rank-consistent, and compatible with isotopic coverage and reference structural information. During refinement, the perceived molecular topology is locked after model formulation; trial steps that would create or remove bonds, including spurious H–H contacts, are rejected, so the coordinate model, the final geometry report, and the uncertainty propagation remain chemically consistent. When the isotopologue set is incomplete, computational reference geometries or curated fragment libraries enter only through explicit constraints, predicates, or descriptor-based parameter classes.

The approach is validated on glycolaldehyde, maleic, phthalic, and succinic anhydrides, cyclopenta-

diene, nitrobenzene, azulene, norcamphor, glycine I/II, and testosterone. These examples span standard acyclic, cyclic, planar aromatic, fused-ring, bridged, homologous anhydride, reduced-dimensionality, and large-molecule cases. Glycine illustrates a complete QC-to-refinement route in which synthon-based diagnostics generate the constraints used by the fit, whereas testosterone is used as a scale demonstration for a large fused-ring system. The results show that the symmetry-adapted GIC and Cartesian active spaces give consistent ranks, effective dimensionalities, and rotational residuals, while final bond lengths, angles, dihedrals, and their uncertainties are obtained from the optimized Cartesian structures with rigorous error propagation. The point is not that a new least-squares engine lowers residuals after the model is built; it is that the same semiexperimental information is obtained without hand-building the coordinate and constraint model.

We believe the manuscript is appropriate for *The Journal of Chemical Physics* because it presents a general computational methodology for spectroscopy-driven molecular structure refinement, not a molecule-specific refinement.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author has approved the submission and declares no competing financial interest. The software, benchmark inputs, generated tables, and regression tests can be made available to reviewers and, upon publication, under the distribution model selected by the author.

Sincerely,

Vincenzo Barone
INSTM, National Interuniversity
Consortium for Materials Science and
Technology
via G. Giusti 9
50121 Firenze, Italy
vincebarone52@gmail.com